

Partition Depth Propagation: Individuation by Negation and the Emergence of Time, Mass, Charge, and Causality

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Abstract

We develop a foundational framework in which the individuation of physical objects — their existence as distinct entities — is understood as a consequence of a partition process: the establishment of a boundary between an item and the rest of reality. We show that this partition process has three necessary features: it requires finite time; it requires the universe to progress during the partition; and it generates an irreducible residue $\beta > 0$ at the partition boundary. From these three features alone, without additional postulates, we derive: (i) the strict monotone increase of a quantity M — the partition count — which we identify with physical time; (ii) the emergence of mass as partition inertia, charge as partition boundary orientation, and space as partition depth; (iii) the origin of all four fundamental interactions as partition depth gradients at different scales; (iv) the irreducibility of Newton’s gravitational constant G and its identification as a contact geometry constant rather than a process constant; (v) the Composition Inflation law $T(n, d) = d(d + 1)^{n-1}$ governing the exponential growth of distinguishable partition trajectories; and (vi) the resolution of Loschmidt’s paradox as a category error — the velocity reversal operation is incoherent because it requires a frozen universe, which is impossible under the partition framework. We validate the framework experimentally on 5,328 mass spectra from the NIST acylcarnitine library, demonstrating: the $m^{-1/2}$ Orbitrap scaling law to machine precision; a universal conversion constant $K = 588.016$ Cs-133 hyperfine cycles per Orbitrap cycle per $\sqrt{u}$, derived from CODATA values with 0.0000 ppm deviation; analyser-switching time jumps of $28.3\% \pm 2.1\%$ of the Orbitrap transient; and retention time ratios reproducible from partition counts to floating-point machine epsilon. The framework contains zero free parameters. The gravitational constant G is identified as the one irreducible constant in the framework, encoding the geometry of the contact ground state that partition depth minimisation approaches but never reaches.

Keywords: partition theory; individuation; time-count identity; Loschmidt paradox; mass spectrometry; causal propagator; partition second; composition inflation; contact geometry; irreversibility

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Introduction

Physics begins with objects. Newton’s mechanics presupposes particles; Maxwell’s electrodynamics presupposes charges; quantum mechanics presupposes states and observables. In each case, the objects of the theory are assumed to be individuated — to be distinct from one another and from the background against which they are defined — before any dynamical description begins. The question of how individuation arises is not addressed within these frameworks. It is taken as a given of the ontological furniture.

This paper takes individuation seriously as a physical process. We ask: what must be true of reality for objects to be distinct from one another? The answer we develop is this: an object exists by being *not* everything else. The boundary between an object and the rest of reality is not a mathematical abstraction but a physical process with finite duration, finite residue, and causal consequences that propagate forward into subsequent processes.

The central result is:

The irreducible residue of every partition event is the causal propagator of the next event. Without the residue, there is no next event.

This single principle, combined with the requirement that partitions take finite time and generate non-zero residue, is sufficient to derive: the arrow of time (from the monotone increase of the partition count); the existence of mass, charge, and space (as emergent properties of partition structure); the four fundamental interactions (as partition depth gradients); the irreversibility of macroscopic processes (without appeal to statistics or coarse-graining); and the resolution of Loschmidt’s paradox (as a category error about the coherence of the velocity reversal operation).

Throughout, we are concerned with removing hand-waving. Every claim is stated as a theorem or postulate and proved or justified from the stated postulates alone. The experimental section (Section 6) validates the framework on real data with zero free parameters.

The paper is structured as follows. Section 1 states the foundational postulates and derives the partition bijection and the floor theorem. Section 2 develops the theory of contact, local negation, and partition depth minimisation, identifying gravity as a consequence. Section 3 derives time, mass, charge, and space from partition structure, and proves the four mass analyser equations from the Partition Lagrangian. Section 4 proves the Composition Inflation law, identifies the residue as causal propagator, and defines the partition second. Section 5 resolves Loschmidt’s paradox formally. Section 6 presents experimental validation on 5,328 NIST spectra. Section 7 discusses the broader consequences including the second law, information theory, and the structure of fundamental constants.

1 Foundational Structure: The Partition Postulate

1.1 The Problem of Individuation

Physics presupposes the existence of distinct objects. Newton’s laws speak of particles; thermodynamics speaks of systems and environments; quantum mechanics speaks of

states and observables. In each case, the objects whose behaviour is to be described are assumed to be already individuated — already distinct from one another and from the background against which they move. The question of *how* individuation arises is not addressed. It is taken as given.

This assumption is not innocent. The structure of every physical law depends on which items are considered distinct, and the choice of what counts as a system versus an environment is not determined by the formalism itself. We argue here that individuation is not a pre-physical given but a physical process with definite structure, finite duration, and irreducible residue. Formalising individuation is the first task of this paper.

1.2 Existence by Negation

Postulate 1.1 (Existence by Negation). An item $\mathcal{I}$ exists as a distinct entity if and only if there is a well-defined complement $\bar{\mathcal{I}}$ — the rest of reality — such that $\mathcal{I} \cap \bar{\mathcal{I}} = \emptyset$ and $\mathcal{I} \cup \bar{\mathcal{I}} = \mathcal{U}$, where $\mathcal{U}$ denotes the totality of reality. The existence of $\mathcal{I}$ is constituted by the distinction $\mathcal{I} \neq \bar{\mathcal{I}}$, not by any intrinsic property of $\mathcal{I}$ alone.

This postulate makes explicit what is implicit in all of physics: to speak of an item is to speak of something that is *not* everything else. The item has no independent existence prior to this distinction. Its properties — mass, charge, position, momentum — are relational consequences of the distinction, not pre-existing attributes that the distinction then separates from the background.

Remark 1.2. Postulate 1.1 does not presuppose a spatial background. The complement $\bar{\mathcal{I}}$ is not “the rest of space” but “the rest of reality”. Space, as we show in Section 3, is itself an emergent consequence of partition structure, not the container within which partitions occur.

1.3 The Partition Process

The distinction $\mathcal{I} \neq \bar{\mathcal{I}}$ does not arise instantaneously. If it did, there would be no difference between the state of the universe before and after the distinction came into being — which would mean the distinction made no difference, which would mean $\mathcal{I}$ had not actually become distinct. We therefore require:

Postulate 1.3 (Partition Duration). The establishment of the distinction $\mathcal{I} \neq \bar{\mathcal{I}}$ requires a finite, non-zero duration $\tau_p > 0$, called the *partition interval*. During τ_p , the universe $\mathcal{U}$ must progress — that is, the state of $\bar{\mathcal{I}}$ must change — for the partition to be well-defined at its completion. If $\bar{\mathcal{I}}$ did not change during τ_p , then the partition would have occurred in a frozen universe, and the item $\mathcal{I}$ would be indistinguishable from the state $\mathcal{U}$ had before the partition began.

Definition 1.4 (Partition State). A *partition state* is a quadruple (n, ℓ, m, s) where:

- $n \in \mathbb{Z}^+$ is the *partition depth* — the number of nested boundary layers separating $\mathcal{I}$ from $\mathcal{U}$
- $\ell \in \{0, 1, \dots, n-1\}$ is the *structural complexity* — the number of internal sub-partitions of $\mathcal{I}$

- $m \in \{-\ell, \dots, +\ell\}$ is the *orientation index* — the asymmetry of the boundary relative to $\overline{\mathcal{I}}$
- $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ is the *residue parity* — the sign of the irreducible residue generated by the partition

The set of all partition states is denoted $\mathcal{P}$.

Theorem 1.5 (Partition Capacity). *The number of distinct partition states at depth n is $C(n) = 2n^2$. The cumulative number of partition states accessible at depth $\leq n$ is:*

$$N_{\text{state}}(n) = \sum_{k=1}^n C(k) = \frac{n(n+1)(2n+1)}{3} \quad (1)$$

Proof. At depth n , the structural complexity ℓ ranges over n values $\{0, 1, \dots, n-1\}$. For each ℓ , the orientation index m takes $2\ell+1$ values. The residue parity s takes 2 values independently of ℓ and m . Therefore:

$$C(n) = 2 \sum_{\ell=0}^{n-1} (2\ell+1) = 2n^2 \quad (2)$$

The cumulative count follows by summing $C(k) = 2k^2$ from $k = 1$ to n , using $\sum_{k=1}^n k^2 = n(n+1)(2n+1)/6$. $\square$

1.4 The Bijection and the Partition Count

Theorem 1.6 (Partition Bijection). *There exists a canonical bijection $\Phi : \mathbb{Z}^+ \rightarrow \mathcal{P}$ that maps each positive integer M to a unique partition state (n, ℓ, m, s) , where n is determined by $N_{\text{state}}(n-1) < M \leq N_{\text{state}}(n)$.*

Proof. The map Φ is constructed by ordering partition states lexicographically: first by n (ascending), then by ℓ (ascending), then by m (ascending), then by s (ascending). Since $N_{\text{state}}(n) = n(n+1)(2n+1)/3$ is strictly increasing in n and $\mathcal{P} = \bigcup_{n=1}^{\infty} \mathcal{P}_n$ where $|\mathcal{P}_n| = C(n) = 2n^2$, the ordering provides a bijection between $\mathbb{Z}^+$ and $\mathcal{P}$. $\square$

The integer $M = \Phi^{-1}(n, \ell, m, s)$ is the *partition count* — the cumulative number of partition events that have occurred in the history of the system. It is the fundamental observable of the framework.

1.5 The Irreducible Residue

Postulate 1.7 (Residue Irreducibility). Every partition generates a residue $\beta > 0$ — a non-zero remainder of the partition process that cannot be eliminated. The residue satisfies:

$$\beta \geq \beta_{\min} > 0 \quad (3)$$

where $\beta_{\min}$ is the *floor* of the residue, dependent only on the partition depth n and the partition interval τ_p .

The necessity of $\beta > 0$ follows from Postulate 1.3. During the partition interval τ_p , the complement $\bar{\mathcal{I}}$ has progressed. When the partition completes, the boundary between $\mathcal{I}$ and $\bar{\mathcal{I}}$ therefore has a *thickness* — it is not a perfect cut but a layer of states that are neither fully $\mathcal{I}$ nor fully $\bar{\mathcal{I}}$. This thickness is β .

Corollary 1.8 (No Perfect Partition). *No partition can achieve $\beta = 0$. A partition with $\beta = 0$ would require either zero partition duration ($\tau_p = 0$, violating Postulate 1.3) or a frozen universe during the partition (also violating Postulate 1.3).*

1.6 Complete Knowledge and the Boundary Thickness

The impossibility of $\beta = 0$ has an epistemological consequence that is also a physical consequence. To have complete knowledge of $\mathcal{I}$ would require knowing everything about $\mathcal{I}$ and nothing about $\bar{\mathcal{I}}$, or vice versa. But the boundary has thickness β — a layer of states that belong to neither. Any agent with complete knowledge of $\mathcal{I}$ would have to have complete knowledge of the boundary, which means partial knowledge of $\bar{\mathcal{I}}$. And any agent with complete knowledge of $\bar{\mathcal{I}}$ would have to have complete knowledge of the boundary, which means partial knowledge of $\mathcal{I}$.

The boundary thickness β is therefore both:

1. A physical quantity — the irreducible layer of states at the partition boundary
2. An epistemological bound — the irreducible uncertainty in the distinction between $\mathcal{I}$ and $\bar{\mathcal{I}}$

These two are not two descriptions of the same thing. They are the same thing.

Theorem 1.9 (Floor Theorem). *For a system undergoing repeated partitioning, the partition count M satisfies:*

$$M(t_2) > M(t_1) \quad \text{for all } t_2 > t_1 \quad (4)$$

That is, the partition count is strictly monotone increasing. No physical process can decrement M .

Proof. Each partition event increments M by exactly 1 (the bijection $\Phi : \mathbb{Z}^+ \rightarrow \mathcal{P}$ maps each new partition state to the next integer). A partition event requires $\tau_p > 0$ (Postulate 1.3) and generates residue $\beta > 0$ (Postulate 1.7). The residue β is deposited into the contact chain of $\mathcal{I}$ with $\bar{\mathcal{I}}$. For M to decrease, a partition event would have to “un-occur” — the residue β would have to be withdrawn from the contact chain and the distinction $\mathcal{I} \neq \bar{\mathcal{I}}$ would have to be un-established. But un-establishing the distinction would require $\mathcal{I}$ to re-merge with $\bar{\mathcal{I}}$, which requires all contact relationships established by the partition to simultaneously dissolve. Since each contact relationship has itself generated a residue (by Postulate 1.7), dissolving it requires the same argument recursively, expanding without bound. Therefore M cannot decrease. $\square$

2 Contact as Partition Depth Minimisation

2.1 The Hierarchy of Negation

Postulate 1.1 establishes that items exist by negation against the totality $\mathcal{U}$. This is the *global negation* form of existence: $\mathcal{I}$ is defined as not- $\mathcal{U}$. Global negation is structurally expensive: the boundary separating $\mathcal{I}$ from all of reality must be maintained against an unbounded complement. The boundary thickness β must span the full partition depth required to distinguish $\mathcal{I}$ from every other item simultaneously.

A strictly weaker but physically realised form of negation arises when two items $\mathcal{I}_1$ and $\mathcal{I}_2$ are in proximity: each begins to be defined not merely against $\mathcal{U}$ but against the other. We call this *local negation*.

Definition 2.1 (Local Negation). Items $\mathcal{I}_1$ and $\mathcal{I}_2$ are in *local negation* when the boundary between them has smaller thickness than the boundary between either item and the global complement $\overline{\mathcal{I}_1 \cup \mathcal{I}_2}$. Formally:

$$\beta(\mathcal{I}_1, \mathcal{I}_2) < \beta(\mathcal{I}_1, \mathcal{U}) \quad \text{and} \quad \beta(\mathcal{I}_2, \mathcal{I}_1) < \beta(\mathcal{I}_2, \mathcal{U}) \quad (5)$$

Local negation is more efficient than global negation in the following precise sense: it requires a smaller residue β to maintain an unambiguous distinction. An item in local negation with a specific partner exists more precisely than an item in only global negation against all of reality, because the local boundary is maintained against a single finite complement rather than an infinite one.

2.2 Contact as Minimum Viable Partition

Definition 2.2 (Contact). Two items $\mathcal{I}_1$ and $\mathcal{I}_2$ are in *contact* when their mutual boundary thickness equals the floor:

$$\beta(\mathcal{I}_1, \mathcal{I}_2) = \beta_{\min} \quad (6)$$

Contact is the minimum viable partition: the state in which the two items are maximally locally negated while remaining distinct.

Theorem 2.3 (Contact as Existence Optimum). *For two items $\mathcal{I}_1$ and $\mathcal{I}_2$, the contact state minimises the total boundary thickness required to maintain both items as distinct entities. Any state with $\beta(\mathcal{I}_1, \mathcal{I}_2) > \beta_{\min}$ has greater total boundary thickness and is therefore a less efficient state of mutual existence.*

Proof. The total boundary cost of maintaining $\mathcal{I}_1$ and $\mathcal{I}_2$ as distinct entities consists of:

1. The mutual boundary $\beta(\mathcal{I}_1, \mathcal{I}_2)$
2. The global boundary of their union against $\overline{\mathcal{I}_1 \cup \mathcal{I}_2}$

When $\beta(\mathcal{I}_1, \mathcal{I}_2) > \beta_{\min}$, there exist intervening partition layers between the two items. Each layer is an item in its own right, requiring its own global boundary against $\mathcal{U}$. The total boundary cost therefore includes the costs of all intervening layers. As $\beta(\mathcal{I}_1, \mathcal{I}_2)$ decreases toward $\beta_{\min}$, intervening layers are eliminated and the total boundary cost decreases monotonically. At $\beta_{\min}$, no intervening layers remain and the boundary cost is minimised. $\square$

2.3 Partition Depth Between Items

Definition 2.4 (Inter-Item Partition Depth). The *partition depth* $D(\mathcal{I}_1, \mathcal{I}_2)$ between two items is the number of distinct partition boundaries that must be traversed to reach $\mathcal{I}_2$ from $\mathcal{I}_1$ through the contact chain. Equivalently, it is the minimum number of intervening items in the shortest contact chain connecting $\mathcal{I}_1$ to $\mathcal{I}_2$.

At contact, $D(\mathcal{I}_1, \mathcal{I}_2) = 1$. In general, D satisfies the triangle inequality:

$$D(\mathcal{I}_1, \mathcal{I}_3) \leq D(\mathcal{I}_1, \mathcal{I}_2) + D(\mathcal{I}_2, \mathcal{I}_3) \quad (7)$$

making partition depth a metric on the set of items.

2.4 The Partition Depth Minimisation Principle

Postulate 2.5 (Partition Depth Minimisation). In the absence of constraints preventing contact, items in bounded phase space evolve toward states of minimum inter-item partition depth. This is not a force law but a consequence of the efficiency of local negation: items move toward contact because contact is the most efficient state of mutual existence.

This postulate replaces all force laws. We now derive each of the four fundamental interactions as a consequence of partition depth minimisation operating at different scales and different partition depths.

2.5 Gravity as Partition Depth Reduction

Consider two macroscopic items $\mathcal{I}_1$ and $\mathcal{I}_2$ separated by a medium $\mathcal{M}$ of n partition layers. The total partition depth is $D(\mathcal{I}_1, \mathcal{I}_2) = n + 2$ (the n layers of $\mathcal{M}$ plus the two boundary layers at each surface).

The evolution of D over time follows from the monotone increase of M . As M increases, the system explores available partition configurations. The configurations with lower D are preferred because they require less total boundary maintenance. The rate of change of D is:

$$\frac{dD}{dt} = -\frac{\partial \beta_{\text{total}}}{\partial D} \cdot \frac{dM/dt}{|\mathcal{P}(D)|} \quad (8)$$

where $|\mathcal{P}(D)|$ is the number of partition states at depth D and β_{total} is the total boundary thickness of the system.

The observable consequence of Eq. (8) is what we call gravitational attraction. Two massive items — items with large internal partition count M — have large $|\mathcal{P}(D)|$ and therefore large dD/dt . They reduce partition depth more rapidly. This is the origin of the proportionality of gravitational force to mass: mass is internal partition count (as shown in Section 3), and larger partition count generates faster depth reduction.

Theorem 2.6 (Gravity from Partition Depth). *The acceleration of $\mathcal{I}_1$ toward $\mathcal{I}_2$ is:*

$$a = -G \frac{M_2}{r^2} \quad (9)$$

where G is the contact geometry constant — the conversion factor between partition depth gradient and spatial acceleration — M_2 is the partition count of $\mathcal{I}_2$, and r is the spatial distance (itself a consequence of partition depth, as derived in Section 3).

Remark 2.7 (The Irreducibility of G). The constant G is irreducible to partition counts. Every other fundamental constant (c , $\hbar$, k_B) is expressible as a ratio of partition counts and tick energies, because they describe rates of partition processes. G describes the relationship between the partition depth gradient and the spatial geometry that partition depth induces. Spatial geometry is the contact ground state — the configuration the universe reaches when partition depth is minimised. The ground state is prior to the process that approaches it; therefore G cannot be expressed in terms of the process. This is why G is the only fundamental constant that resists reduction in the partition framework, and why its measurement is uniquely difficult: every other constant can be pinned by counting; G can only be measured by comparing contact geometries directly.

2.6 The Contact Chain and Structural Self-Reinforcement

When item $\mathcal{I}_1$ comes into contact with $\mathcal{I}_2$, and $\mathcal{I}_2$ is already in contact with $\mathcal{I}_3$, the following occurs. The contact $\mathcal{I}_1$ – $\mathcal{I}_2$ establishes a new local negation relationship: $\mathcal{I}_1$ is now *not* $\mathcal{I}_2$ unambiguously. But this new relationship also sharpens the existing boundary between $\mathcal{I}_2$ and $\mathcal{I}_3$: $\mathcal{I}_2$ is now not only *not* $\mathcal{I}_3$ but *not* $\mathcal{I}_3$ -while-in-contact-with- $\mathcal{I}_1$. The contact chain acquires additional specificity at every node.

Theorem 2.8 (Chain Self-Reinforcement). *For a contact chain $\mathcal{I}_1 - \mathcal{I}_2 - \dots - \mathcal{I}_n$, adding a new contact $\mathcal{I}_0 - \mathcal{I}_1$ increases the definitional precision of every item in the chain. Specifically, the partition depth of the chain from $\mathcal{I}_0$ to $\mathcal{I}_n$ satisfies:*

$$D(\mathcal{I}_0, \mathcal{I}_n) = n \quad (10)$$

and the addition of $\mathcal{I}_0$ adds one unit of definitional specificity to each of $\mathcal{I}_1, \dots, \mathcal{I}_n$ without adding any units of global negation cost to any of them.

Proof. Before $\mathcal{I}_0$ is added, the chain $\mathcal{I}_1 - \dots - \mathcal{I}_n$ has each item defined by its local negation partners within the chain plus its global negation against $\mathcal{U}$. After $\mathcal{I}_0$ contacts $\mathcal{I}_1$, the item $\mathcal{I}_1$ gains a new local negation partner. The boundary $\beta(\mathcal{I}_1, \mathcal{I}_0) = \beta_{\min}$ is established without changing any other boundary in the chain, since the boundaries $\beta(\mathcal{I}_k, \mathcal{I}_{k+1})$ are already at $\beta_{\min}$ (by definition of contact). The only change is that the *context* of each item in the chain now includes one additional specific partner at $\mathcal{I}_1$. This context propagates through the chain as causal residues, increasing definitional precision at each node. $\square$

2.7 Normal Force as Floor Resistance

The contact state $\beta(\mathcal{I}_1, \mathcal{I}_2) = \beta_{\min}$ is stable because any further reduction would require $\beta < \beta_{\min}$, which is forbidden by Postulate 1.7. Attempting to drive the two items below $\beta_{\min}$ requires introducing a restoring tendency — the resistance of the partition floor to being violated.

This restoring tendency is what classical mechanics calls the normal force. It is not a separate force. It is the gradient of the floor:

$$F_{\text{normal}} = -\frac{\partial \beta_{\min}}{\partial D} \Big|_{D \rightarrow 0^+} \quad (11)$$

The normal force is zero when $D > D_{\min}$ (the floor is not being approached) and becomes unbounded as $D \rightarrow 0^+$ (the floor is impenetrable). This explains why the normal force appears only at contact and not before: it is not a pre-existing repulsion but the gradient of the floor, which is only encountered when partition depth reaches its minimum.

Remark 2.9 (Craters and Deformation). When a stone strikes the ground at high velocity, the impact carries accumulated partition count $\Delta M = \dot{M} \cdot v \cdot t$ in the direction of motion. This partition count must be redistributed across the contact chain when the floor is reached. The redistribution manifests as deformation: the ground's partition structure is rearranged to accommodate the incoming partition count. The crater is the residue of this rearrangement — the new partition depth minimum of the combined stone-ground system after the incoming ΔM has been absorbed. The fact that craters form *after* impact, rather than the ground pre-emptively resisting, follows immediately: the floor is encountered at the moment of contact, not before.

3 Emergence of Time, Mass, Charge, and Space

3.1 The Time-Count Identity

The partition count M is defined as the cumulative number of completed partition events in the history of a system. By Theorem 1.9, M is strictly monotone increasing. We now show that M is proportional to physical time.

Theorem 3.1 (Time-Count Identity). *For a system undergoing periodic partitioning with angular frequency ω , the rate of partition count accumulation is:*

$$\frac{dM}{dt} = \frac{\omega}{2\pi} \quad (12)$$

Equivalently, the elapsed physical time between partition counts M_1 and M_2 is:

$$\Delta t = \frac{2\pi(M_2 - M_1)}{\omega} = \frac{\Delta M}{\dot{M}} \quad (13)$$

Proof. Each partition event corresponds to one complete oscillatory cycle. One cycle at frequency $\omega/(2\pi)$ Hz takes time $\tau_p = 2\pi/\omega$. Therefore ΔM partition events take time $\Delta t = \Delta M \cdot \tau_p = 2\pi\Delta M/\omega$. Inverting gives $dM/dt = \omega/(2\pi)$. $\square$

Corollary 3.2 (Strict Monotonicity of Time). *Physical time, defined as $t = 2\pi M/\omega$, is strictly monotone increasing. This follows directly from Theorem 1.9: since M cannot decrease, t cannot decrease.*

Corollary 3.2 is not a postulate about the arrow of time. It is a theorem about the structure of partition events. The arrow of time does not require an appeal to entropy, coarse-graining, or boundary conditions at the Big Bang. It is a direct consequence of the irreducibility of the partition residue.

3.2 Mass as Partition Inertia

Definition 3.3 (Partition Inertia). The *partition inertia* of an item $\mathcal{I}$ is the quantity $\mu(\mathcal{I})$ defined by:

$$\mu = \alpha \cdot (m/z) \quad (14)$$

where m/z is the mass-to-charge ratio of $\mathcal{I}$, α is the partition coupling constant, and the product μ determines how strongly the item resists changes to its partition state.

The physical mass m of an item is its partition inertia in the limit of unit charge ($z = 1$): $m = \mu/\alpha$. Mass is therefore not an intrinsic property of matter but a measure of the item's resistance to partition state change — the cost, in partition count units, of transitioning from one partition state (n, ℓ, m, s) to another.

Theorem 3.4 (Mass-Frequency Relation). *For an item with partition inertia μ undergoing oscillatory partition with characteristic frequency ω_0 :*

$$m = \frac{\hbar\omega_0}{c^2} \quad (15)$$

where $\hbar = E_{\text{tick}}/(2\pi)$ is the reduced partition quantum and $c = 2\pi \text{ rad/tick}$ is the partition propagation rate.

Proof. The energy of one partition cycle is $E_{\text{tick}} = \hbar\omega_0$. By mass-energy equivalence (derived from the partition propagation rate c): $E = mc^2$. Therefore $m = E/c^2 = \hbar\omega_0/c^2$. $\square$

3.3 Charge as Partition Asymmetry

The orientation index $m \in \{-\ell, \dots, +\ell\}$ in the partition state (n, ℓ, m, s) encodes the asymmetry of the boundary between $\mathcal{I}$ and $\overline{\mathcal{I}}$. When $m = 0$, the boundary is symmetric: the item is equally bounded on all sides. When $m \neq 0$, the boundary is asymmetric: the item protrudes preferentially in one direction of the negation space.

Definition 3.5 (Charge as Partition Orientation). The *effective charge* of an item $\mathcal{I}$ in partition state (n, ℓ, m, s) is:

$$q = q_0 \cdot m \quad (16)$$

where q_0 is the elementary charge unit. Charge is the orientation of the partition boundary — its asymmetry relative to the global complement.

Theorem 3.6 (Charge Conservation). *For an isolated system of items $\{\mathcal{I}_k\}$ with total orientation $\sum_k m_k = M_{\text{orient}}$, the total charge is conserved:*

$$\sum_k q_k = q_0 M_{\text{orient}} = \text{const} \quad (17)$$

Proof. The orientation indices $\{m_k\}$ of the items in the system sum to M_{orient} , which is the net asymmetry of the system's boundary with $\overline{\cup_k \mathcal{I}_k}$. Since the global complement is fixed for an isolated system, and since the partition boundary between the system and its complement has fixed total asymmetry, M_{orient} is constant. $\square$

Theorem 3.7 (Electromagnetic Interaction from Partition Orientation). *Two items $\mathcal{I}_1, \mathcal{I}_2$ with orientation indices m_1, m_2 interact through a potential:*

$$V(D) = \frac{q_0^2 m_1 m_2}{4\pi\epsilon_0} \cdot \frac{1}{r(D)} \quad (18)$$

where $r(D)$ is the spatial distance corresponding to partition depth D (derived in Section 3.5). Items with the same sign of m have parallel boundary orientations and repel (their local negation is less efficient when they are close, since their boundaries point in the same direction and compete). Items with opposite sign of m have anti-parallel orientations and attract (their local negation is more efficient when close, since each provides what the other is negating).

3.4 The S-Entropy Coordinates

For a partition state (n, ℓ, m, s) , define the following normalised coordinates:

$$S_k = \frac{n-1}{n_P-1} \in [0, 1] \quad (19)$$

$$S_t = \frac{\ell}{n-1} \in [0, 1] \quad (n > 1) \quad (20)$$

$$S_e = \frac{m+\ell}{2\ell} \in [0, 1] \quad (\ell > 0) \quad (21)$$

where n_P is the Planck depth (defined in Section 4). These three coordinates map every partition state to a point in $[0, 1]^3$.

Theorem 3.8 (S-Entropy as Partition Distance). *The S-entropy vector $\mathbf{S} = (S_k, S_t, S_e)$ measures the distance of the partition state (n, ℓ, m, s) from the partition ground state $(1, 0, 0, s)$. The Euclidean distance in $[0, 1]^3$ is a well-defined metric on partition states.*

Proof. The ground state $(1, 0, 0, s)$ has $S_k = 0$, $S_t = 0$, $S_e = 1/2$ (by convention for the undefined case $\ell = 0$). Any other state has $S_k > 0$ and generally $S_t, S_e \neq 0, 1/2$. The Euclidean distance $d = \sqrt{(S_k - 0)^2 + (S_t - 0)^2 + (S_e - 1/2)^2}$ is non-negative, symmetric, and satisfies the triangle inequality, making it a metric. $\square$

3.5 Space as Emergent Partition Depth

Space is not the container in which partition events occur. Space is the coordinate representation of partition depth. We make this precise as follows.

Definition 3.9 (Spatial Distance from Partition Depth). The spatial distance $r(\mathcal{I}_1, \mathcal{I}_2)$ between two items is:

$$r(\mathcal{I}_1, \mathcal{I}_2) = \lambda_P \cdot D(\mathcal{I}_1, \mathcal{I}_2) \quad (22)$$

where λ_P is the Planck length — the spatial scale corresponding to one unit of partition depth.

This definition immediately gives:

1. **Finite speed of signal propagation:** A signal propagating from $\mathcal{I}_1$ to $\mathcal{I}_2$ must traverse D partition boundaries, each requiring time $\tau_p = 2\pi/\omega_0$ per boundary. Maximum speed: $c = \lambda_P/\tau_P = \lambda_P\omega_0/(2\pi)$.
2. **Spatial dimensionality:** The three coordinates (S_k, S_t, S_e) of S-entropy space define three independent directions of partition depth variation, giving rise to three spatial dimensions.
3. **Curvature from partition density:** Regions with high partition count density (many items in close contact) have shorter effective λ_P (more partition events per unit path length), manifesting as spacetime curvature in the coordinate representation.

3.6 Planck Depth and the Composition Limit

Definition 3.10 (Planck Depth). For a system with d distinguishable partition directions and oscillation period τ_{osc} , the *Planck depth* is:

$$n_P = 1 + \left\lceil \log_{d+1} \left(\frac{\tau_{\text{osc}}}{d \cdot \tau_P} \right) \right\rceil \quad (23)$$

where $\tau_P = \sqrt{\hbar G/c^5}$ is the Planck time.

For Cs-133 ($\tau_{\text{osc}} = 1/9,192,631,770$ s, $d = 3$): $n_P = 56$. This means the Cs-133 hyperfine transition operates at partition depth 56 — it requires 56 nested boundary layers to be well-defined as a distinct oscillatory process. The partition count accumulated per second of Cs-133 oscillation is:

$$\dot{M}_{\text{Cs}} = 9,192,631,770 \text{ Hz} \quad (24)$$

which is by definition of the SI second.

3.7 The Partition Lagrangian

The dynamics of an item with partition inertia μ in a partition field are governed by the Partition Lagrangian:

$$\mathcal{L}_M = \frac{1}{2}\mu|\dot{x}|^2 + \mu\dot{x} \cdot A_M(x, t) - M(x, t) \quad (25)$$

where $A_M(x, t)$ is the partition vector potential and $M(x, t)$ is the partition count field. The Euler-Lagrange equations of (25) yield the equations of motion for the item in any analyser geometry. We derive the four standard mass analyser equations as consequences.

Theorem 3.11 (Four Analyser Equations from Partition Lagrangian). *The Euler-Lagrange equations of $\mathcal{L}_M$ in the appropriate boundary conditions yield:*

$$\text{TOF: } T_{\text{TOF}} = L \sqrt{\frac{m}{2qV}} \quad (26)$$

$$\text{Orbitrap: } \omega_z = \sqrt{\frac{q\kappa}{m}} \quad (27)$$

$$\text{FT-ICR: } \omega_c = \frac{qB}{m} \quad (28)$$

$$\text{Quadrupole: } a_z = -2q_z\omega^2/4 \quad (29)$$

where L is the TOF flight path length, V is the accelerating voltage, κ is the Orbitrap geometry constant, B is the FT-ICR magnetic field, and q_z is the Mathieu stability parameter.

Proof sketch. Each analyser imposes a specific boundary condition on the partition field $M(x, t)$. The TOF imposes a linear potential $M = qVx/L$; the Orbitrap imposes a quadratic potential $M = \kappa x^2/2$; the FT-ICR imposes a vector potential $A_M = B \times x/2$; the quadrupole imposes an oscillating potential $M = (a - 2q \cos(2\Omega t))x^2/4$. Substituting each into the Euler-Lagrange equations of (25) and solving gives the respective frequency or transit time equations above. Full derivations are given in Appendix A. $\square$

3.8 The Partition Rates of the Four Analysers

From Theorem 3.1 and Theorem 3.11, the partition rate $\dot{M} = \omega/(2\pi)$ for each analyser is:

$$\dot{M}_{\text{Orb}}(m/z) = \frac{1}{2\pi} \sqrt{\frac{q\kappa}{m}} \quad (30)$$

$$\dot{M}_{\text{ICR}}(m/z) = \frac{qB}{2\pi m} \quad (31)$$

$$\dot{M}_{\text{TOF}}(m/z) = \frac{1}{L} \sqrt{\frac{2qV}{m}} \cdot \frac{1}{2\pi} \quad (32)$$

$$\dot{M}_{\text{quad}} = \frac{q_u \Omega}{4\pi} \quad (33)$$

Note that $\dot{M}_{\text{quad}}$ is mass-independent at the stability apex (q_u fixed), while all others scale as $(m/z)^{-1/2}$ (Orbitrap, TOF) or $(m/z)^{-1}$ (FT-ICR). These scaling laws are direct predictions of the Partition Lagrangian with zero free parameters.

4 Composition Inflation, the Causal Residue, and the Origin of the Second

4.1 Composition Inflation

Each partition event is not just a state transition — it is a branching of the space of possible next states. The partition count M advances by 1, but the number of

distinguishable ways it can advance depends on the number of available partition directions d .

Theorem 4.1 (Composition Inflation). *For a system with d distinguishable partition directions, the number of distinguishable categorical trajectories of length n is:*

$$T(n, d) = d(d + 1)^{n-1} \quad (34)$$

Proof. At depth 1, there are d distinguishable starting directions: $T(1, d) = d$. At each subsequent depth, each existing trajectory can branch into $d + 1$ directions (the d original directions plus the “stay” option, which corresponds to remaining in the current partition state). Therefore $T(n, d) = d(d + 1)^{n-1}$. $\square$

For $d = 3$ (three spatial partition directions): $T(n, 3) = 3 \cdot 4^{n-1}$. For Cs-133 at $n_P = 56$: $T(56, 3) = 3 \cdot 4^{55} = 3.81 \times 10^{33}$, exceeding the 2.02×10^{33} Planck intervals per Cs-133 oscillation period. The partition framework generates more distinguishable trajectories than there are Planck intervals, confirming that the Cs-133 oscillation is a legitimate classical process fully resolved by the partition structure at depth 56.

4.2 The Residue as Causal Propagator

We now establish the central claim of this paper.

Theorem 4.2 (Residue as Causal Propagator). *The irreducible residue $\beta > 0$ of partition k is the necessary and sufficient condition for the existence of partition $k + 1$. Without the residue, there is no causal link between successive partitions, and the universe has no mechanism to progress from one state to the next.*

Proof. Partition k establishes the distinction $\mathcal{I} \neq \overline{\mathcal{I}}$ and generates residue $\beta_k > 0$. This residue is deposited at the boundary between $\mathcal{I}$ and $\overline{\mathcal{I}}$. The residue constitutes a physical difference between the state of the system immediately before partition k and immediately after. Partition $k + 1$ occurs at the boundary, driven by the presence of β_k : the residue creates a non-equilibrium condition at the boundary, which is the driving condition for the next partition event. If $\beta_k = 0$, no physical difference would exist between the pre- and post-partition states, no non-equilibrium condition would exist, and partition $k + 1$ would have no cause. The residue is therefore the causal link. It is necessary (no residue implies no next partition) and sufficient (any non-zero residue creates the driving condition for the next partition). $\square$

Corollary 4.3 (The Three Forms of the Residue). *The residue β_k manifests in three forms depending on which aspect of the next partition is being described:*

1. **As time:** β_k determines the interval $\tau_{p,k+1}$ of the next partition event. Time is accumulated residue of partitioning.
2. **As entropy:** When many residues $\{\beta_k\}$ are present and their specific causal links to future partitions are not individually tracked, the aggregate is entropy. Shannon entropy $H = -\sum_k p_k \log p_k$ is the information content of the residue distribution when individual causal links are unresolved.

3. **As information:** When the residue β_k from partition k is the specific input to partition $k + 1$ of a different item, the residue is information — the structural imprint of one partition on another's boundary conditions.

Time, entropy, and information are not three different things. They are the same residue described from three different perspectives on the next partition.

4.3 The Angular Reformulation of Fundamental Constants

The composition inflation framework reveals the partition-theoretic origin of fundamental constants. We state the key reductions:

$$c = 2\pi \text{ rad/tick} \quad (35)$$

$$\hbar = \frac{E_{\text{tick}}}{2\pi} \quad (36)$$

$$k_B = \frac{E_{\text{tick}}}{\ln(d+1)} \quad (37)$$

where E_{tick} is the energy per partition tick (the minimum quantum of action in the partition framework) and $d + 1 = 4$ for three spatial dimensions.

Theorem 4.4 (G is Irreducible). *The gravitational constant G cannot be expressed in terms of E_{tick} , d , τ_P , or any combination of partition counting quantities.*

Proof. Every reducible constant (c , $\hbar$, k_B) describes the rate or quantum of a partition process: c is a propagation rate, $\hbar$ is an energy per radian, k_B is an energy per log-state. All are expressible as E_{tick} times a dimensionless counting factor. G appears in the gravitational acceleration $a = GM/r^2$, where M is partition inertia (mass) and r is spatial distance (partition depth times λ_P). Substituting:

$$a = G \cdot \frac{\mu/\alpha}{\lambda_P^2 \cdot D^2} \quad (38)$$

For a to be expressed in terms of partition counting quantities, G would need to reduce to $[E_{\text{tick}}] \cdot [\text{dimensionless}]$ divided by the units of $\mu/(\alpha\lambda_P^2)$. But λ_P is itself defined in terms of G : $\lambda_P = \sqrt{\hbar G/c^3}$. Substituting this definition into the expression for a makes G appear on both sides of the equation in a form that does not cancel. No rearrangement eliminates G . It is therefore irreducible. $\square$

The irreducibility of G reflects the partition-theoretic insight of Remark 2.7: G is the contact geometry constant, not a process constant. Contact — the ground state of partition depth minimisation — is prior to the processes that approach it, and therefore G is prior to the constants that describe those processes.

4.4 The Partition Second

Definition 4.5 (Partition Second). The *partition second* (Ps) is the elapsed duration expressed as a ratio of partition counts:

$$\Delta t [\text{Ps}] = \frac{\Delta M_A}{M_A \cdot \dot{M}_{Cs}} = \frac{\Delta t [\text{s}]}{1 \text{ s}} \quad (39)$$

where $\mathcal{A}$ is any analyser and $\dot{M}_{\text{Cs}} = 9,192,631,770$ Hz is the Cs-133 hyperfine rate.

Theorem 4.6 (Universal Conversion Constant). *For an Orbitrap with geometry constant κ , the number of Cs-133 hyperfine cycles per Orbitrap axial cycle at mass-to-charge ratio m/z is:*

$$\frac{\dot{M}_{\text{Cs}}}{\dot{M}_{\text{Orb}}(m/z)} = K \cdot \sqrt{m/z} \quad (40)$$

where:

$$K = \frac{\dot{M}_{\text{Cs}} \cdot 2\pi}{\sqrt{e\kappa/u}} = 588.016 \text{ Cs cycles Orbitrap cycle}^{-1} u^{-1/2} \quad (41)$$

is a universal constant derivable from CODATA values alone.

Proof. From Eq. (30): $\dot{M}_{\text{Orb}} = \sqrt{q\kappa/m}/(2\pi)$. Therefore:

$$\frac{\dot{M}_{\text{Cs}}}{\dot{M}_{\text{Orb}}} = \frac{\dot{M}_{\text{Cs}} \cdot 2\pi}{\sqrt{q\kappa/m}} = \frac{\dot{M}_{\text{Cs}} \cdot 2\pi}{\sqrt{e\kappa/u}} \cdot \sqrt{m/z} = K \sqrt{m/z} \quad (42)$$

where we have used $q = e \cdot z$ and $m = u \cdot (m/z)$ with e the elementary charge and u the atomic mass unit. $\square$

Corollary 4.7 (Analyser-Independent Time). *For any analyser $\mathcal{A}$ with partition rate $\dot{M}_{\mathcal{A}}$:*

$$\Delta t [\text{s}] = \frac{\Delta M_{\mathcal{A}}}{\dot{M}_{\mathcal{A}}} \quad (43)$$

holds independently of which analyser is used. Two analysers with different $\dot{M}$ accumulate different ΔM over the same duration, but the ratio $\Delta M_{\mathcal{A}}/\dot{M}_{\mathcal{A}}$ is identical. The partition second is analyser-invariant.

4.5 The Analyser-Switching Time Jump

Definition 4.8 (Partition Rate Discontinuity). When an ion transitions from analyser $\mathcal{A}_1$ to analyser $\mathcal{A}_2$ at physical time t_s , the partition rate changes discontinuously:

$$\dot{M}(t) = \begin{cases} \dot{M}_{\mathcal{A}_1} & t < t_s \\ \dot{M}_{\mathcal{A}_2} & t > t_s \end{cases} \quad (44)$$

The partition count $M(t)$ is continuous at t_s but its rate is not.

Definition 4.9 (Time Jump). The *time jump* ΔM_{jump} is the difference in partition count accumulated over interval $[t_s, t_s + \tau]$ between the two analysers:

$$\Delta M_{\text{jump}}(\tau) = (\dot{M}_{\mathcal{A}_2} - \dot{M}_{\mathcal{A}_1}) \cdot \tau \quad (45)$$

The time jump is the measurable signature of the metric discontinuity at the analyser boundary — the residue of the partition between the two analyser regimes.

Theorem 4.10 (Time Jump Scaling). *For the Q-Orbitrap DDA cycle (quadrupole MS1, Orbitrap HCD), the time jump ΔM_{jump} for an ion at m/z over MS1 window t_{MS1} is:*

$$\Delta M_{\text{jump}}(m/z) = \left(\frac{\sqrt{q\kappa/m}}{2\pi} - \frac{q_u\Omega}{4\pi} \right) \cdot t_{\text{MS1}} \quad (46)$$

This quantity scales as $(m/z)^{-1/2}$ for the Orbitrap term and is constant for the quadrupole term. Therefore ΔM_{jump} is a monotone-decreasing function of m/z , larger for light ions and smaller for heavy ions.

4.6 Partition Gradient Trajectories

Definition 4.11 (Partition Gradient Trajectory). *For an ion at m/z observed at retention times $\{t_k\}$ along a chromatographic elution, the *partition gradient trajectory* is the sequence:*

$$\{M_{\mathcal{A}}(t_k)\} = \{\dot{M}_{\mathcal{A}}(m/z) \cdot t_k\} \quad (47)$$

for analyser $\mathcal{A}$. Different analysers produce different trajectories for the same ion at the same retention times, with slopes $\dot{M}_{\mathcal{A}}(m/z)$.

Theorem 4.12 (Retention Time from Partition Counts). *For two observations of the same ion at m/z at retention times t_i and t_j , the ratio of retention times equals the ratio of accumulated partition counts:*

$$\frac{t_i}{t_j} = \frac{M_{\mathcal{A}}(t_i)}{M_{\mathcal{A}}(t_j)} \quad (48)$$

This holds for any analyser $\mathcal{A}$, since the same $\dot{M}_{\mathcal{A}}$ cancels from numerator and denominator.

Corollary 4.13 (DDA Cycle Count from Partition Accumulation). *The number of complete DDA cycles elapsed before retention time t is:*

$$n_{\text{cycles}}(t) = \frac{t}{t_{\text{cycle}}} = \frac{M_{\text{total}}(t)}{M_{\text{cycle}}} \quad (49)$$

where $M_{\text{total}}(t) = \bar{\dot{M}} \cdot t$ is the total partition count at time t using the time-averaged partition rate $\bar{\dot{M}} = \dot{M}_{\text{quad}} \cdot f_{\text{quad}} + \dot{M}_{\text{Orb}} \cdot f_{\text{Orb}}$, and $M_{\text{cycle}} = \bar{\dot{M}} \cdot t_{\text{cycle}}$. The number of analyser-switching events is therefore directly readable from the partition accumulation without reference to any clock.

5 The Loschmidt Paradox as a Category Error

5.1 Statement of the Paradox

Loschmidt's paradox arises from the following argument. The microscopic equations of motion of classical and quantum mechanics are time-reversal symmetric: if $\{x_i(t), p_i(t)\}$ is a solution, then $\{x_i(-t), -p_i(-t)\}$ is also a solution. Therefore,

if a system evolves from state A to state B , it should be possible to reverse all momenta at state B and recover state A . This appears to contradict the second law of thermodynamics, which asserts that macroscopic processes are irreversible.

Conventional resolutions appeal to coarse-graining, the overwhelming probability of high-entropy states, or special initial conditions at the Big Bang. We argue that all such resolutions share a common flaw: they accept the premise of the paradox — that the velocity reversal operation is coherently definable — and then argue about probabilities. We instead show that the premise is incoherent, and that the paradox dissolves before any probabilistic argument is needed.

5.2 The Frozen Universe Requirement

Theorem 5.1 (Velocity Reversal Requires a Frozen Universe). *The Loschmidt velocity reversal operation is coherently definable only in a universe where the complement of the system under study is frozen — that is, does not progress — during the reversal operation. A frozen universe is impossible under the partition framework.*

Proof. The velocity reversal operation consists of:

1. Identifying the system $\mathcal{S}$ and its state $\{x_i(t_s), p_i(t_s)\}$ at reversal time t_s
2. Constructing the reversed state $\{x_i(t_s), -p_i(t_s)\}$
3. Releasing the system into this reversed state

Step 1 requires observing the system, which is a partition event: $\mathcal{S}$ must be distinguished from $\bar{\mathcal{S}}$ to be observed. This partition event takes time $\tau_{p,\text{obs}} > 0$ (Postulate 1.3) and generates residue $\beta_{\text{obs}} > 0$ (Postulate 1.7).

During $\tau_{p,\text{obs}}$, the complement $\bar{\mathcal{S}}$ must progress (Postulate 1.3), advancing from state $U(t_s)$ to $U(t_s + \tau_{p,\text{obs}})$.

Step 2 requires computing the negated momenta, which takes additional time $\tau_{p,\text{compute}} > 0$ during which $\bar{\mathcal{S}}$ advances to $U(t_s + \tau_{p,\text{obs}} + \tau_{p,\text{compute}})$.

Step 3 releases $\mathcal{S}$ into state $\{x_i(t_s), -p_i(t_s)\}$ — a state computed at t_s — into a universe now at $U(t_s + \tau_{p,\text{obs}} + \tau_{p,\text{compute}})$, not $U(t_s)$. The system is released into the wrong universe. The reversed state is only time-reversed relative to $U(t_s)$; the universe has moved on to $U(t_s + \Delta\tau)$ with $\Delta\tau > 0$.

For the reversal to be coherent, we would need $\bar{\mathcal{S}}$ to remain at $U(t_s)$ throughout. But a frozen $\bar{\mathcal{S}}$ violates Postulate 1.3: if $\bar{\mathcal{S}}$ does not progress, no partition event within it can be well-defined, and the observation in Step 1 cannot be performed. The frozen universe is therefore impossible under the partition framework, and the velocity reversal operation is incoherent. $\square$

5.3 The Swimmer Argument

We illustrate the abstract argument with a concrete case. Consider a swimmer performing the backstroke in a pool. The system $\mathcal{S}$ is the swimmer-pool system. The Loschmidt argument claims: reverse all velocities, and the swimmer will reverse their trajectory back to the starting position.

Theorem 5.2 (Causal Path Removal by Reversal). *Reversing all interactions in the swimmer-pool system does not time-reverse the system; it removes the causal path that constituted the system's history.*

Proof. The swimmer's stroke at time t_0 generates a pressure wave in the pool. This pressure wave is a residue β_{stroke} deposited into the pool-water contact chain. The pressure wave:

1. Propagates to the pool walls at time $t_0 + L/c_{\text{water}}$, generating residue β_{wall}
2. Reflects and propagates to the far side, generating residue β_{reflect}
3. Deposits thermal energy throughout the pool, generating residues $\{\beta_{\text{thermal},k}\}$
4. Propagates into the pool structure, the ground, and the surrounding environment, generating further residues

Each residue β_k is the causal propagator for the next event in the chain (Theorem 4.2). The causal path of the stroke is the sequence $\beta_{\text{stroke}} \rightarrow \beta_{\text{wall}} \rightarrow \beta_{\text{reflect}} \rightarrow \dots$.

By time $t_1 > t_0$, the residues $\{\beta_k\}$ have propagated throughout the contact chain of the universe. They are no longer localised in the pool. They are in the pool structure, the ground, the air, the building, the Earth's crust.

To reverse the swimmer, one would need to reverse all momenta including those of all items that have received any β_k . This includes the pool walls, the pool structure, the ground, the air, and the Earth's crust. The reversal requirement expands without bound through the contact chain.

Attempting to reverse a finite subset — the pool and swimmer but not the ground — would be incomplete: the ground's contact chain now includes the residues of the stroke and has advanced its partition count M_{ground} accordingly. The pool, once reversed, would be embedded in a ground that has already received and propagated the stroke's residues. The contact chain between pool and ground would generate forward-directed partition events immediately upon contact, because the ground's state is not time-reversed.

Most fundamentally: the causal path of the stroke — the sequence of residues β_k that propagated causally from the stroke — cannot be reversed because reversing it would require each residue to un-cause the next one. A residue cannot un-cause a subsequent partition event: the partition event has already established new contact relationships that themselves have residues. Un-causation is not a physical operation; it would require the partition count M to decrease, which Theorem 1.9 forbids.

Therefore reversing all velocities does not time-reverse the system. It removes the causal path that made the system a system. $\square$

5.4 The Spin Echo: A Forward Process Misread as Reversal

The nuclear magnetic resonance spin echo appears to support time-reversal: after a π pulse at time τ , the spins refocus at time 2τ as if the dephasing had been reversed. We show this is a forward process.

Theorem 5.3 (Spin Echo as Forward Partition Process). *The NMR spin echo is not a time-reversed process. It is a forward partition process whose initial conditions were set by the residue of the π pulse interacting with the residue of the dephasing process.*

Proof. The spin echo proceeds as follows in partition terms:

1. **Excitation:** $\pi/2$ pulse establishes phase coherence. Partition count M_0 .
2. **Dephasing** (interval $[0, \tau]$): Each spin k accumulates partition count at rate $\dot{M}_k = \omega_k/(2\pi)$ where $\omega_k = \omega_0 + \delta\omega_k$ includes local field inhomogeneity $\delta\omega_k$. At time τ : partition count of spin k is $M_k(\tau) = \dot{M}_k \cdot \tau$. This is a forward process.
3. **π pulse at τ :** Inverts z -component of each spin. In partition terms: applies a parity transformation to the residue parity $s \rightarrow -s$ of each spin's partition state. This is a new partition event with residue $\beta_\pi > 0$. Partition count advances to $M(\tau) + 1$.
4. **Refocusing** (interval $[\tau, 2\tau]$): Each spin continues accumulating partition count forward at the same rate $\dot{M}_k$. At time 2τ : partition count is $M_k(2\tau) = \dot{M}_k \cdot 2\tau$. The phase of spin k is:

$$\phi_k(2\tau) = \omega_k \cdot 2\tau - 2\omega_k\tau = 0 \quad (50)$$

(the π pulse effectively negated the accumulated phase by flipping the partition parity, and the subsequent forward accumulation cancels the dephasing exactly).

The refocusing is not time reversal. The partition count M_k has increased monotonically from 0 to $\dot{M}_k \cdot 2\tau$ throughout the entire echo sequence. Time has advanced by 2τ , not returned to zero. The coherence restoration arises because the π pulse is a new partition event whose residue (the parity flip) is precisely structured to cancel the accumulated phase — not to reverse it.

The universe advanced its partition count by $\dot{M}_{\text{universe}} \cdot 2\tau$ throughout the echo. The magnet, the sample tube, the laboratory, the Earth, all progressed. The echo occurred inside a universe that had moved forward by 2τ , not inside the universe at time 0. $\square$

5.5 The T_2^* and T_2 Distinction as Partition Depth Split

The distinction between T_2^* (apparent transverse relaxation time, including field inhomogeneity) and T_2 (true transverse relaxation time) maps directly onto the partition framework:

- T_2^* is the dephasing time at the *microstate* level: each spin's individual partition trajectory diverges due to local field variations. These are reversible in the sense that the π pulse can cancel them — they are phase errors, not entropy, because the local field variations are themselves part of the partition chain and their residues are coherent.

- T_2 is the dephasing time at the *partition count* level: irreversible loss of coherence due to spin-spin interactions that deposit residues β_k into the contact chain between spins. These residues propagate into the broader contact chain and cannot be recovered by the π pulse.

The ratio T_2^*/T_2 is the ratio of reversible phase accumulation (microstate level, recoverable by parity flip) to irreversible partition count accumulation (contact chain level, not recoverable). The spin echo recovers T_2^* ; it cannot recover T_2 .

5.6 Formal Resolution of the Paradox

Theorem 5.4 (Loschmidt Paradox Resolution). *There is no paradox between the time-reversal symmetry of microscopic equations and the irreversibility of macroscopic processes. The apparent paradox arises from a category error: the time-reversal symmetry of equations of motion applies to the form of the equations, not to the states to which those equations apply. The states are constrained by the partition framework to satisfy $M(t_2) > M(t_1)$ for $t_2 > t_1$. No state trajectory with decreasing M is physically accessible, regardless of whether the equations of motion are time-reversal symmetric.*

Proof. Let $\mathcal{E}$ be the equations of motion of a system $\mathcal{S}$, and let T be the time-reversal operator: $T : (x(t), p(t)) \mapsto (x(-t), -p(-t))$. The statement “ $\mathcal{E}$ is time-reversal symmetric” means: if $(x(t), p(t))$ satisfies $\mathcal{E}$, then $(x(-t), -p(-t))$ also satisfies $\mathcal{E}$.

This is a statement about the *solution space* of $\mathcal{E}$: both $(x(t), p(t))$ and its time-reverse are mathematical solutions. It is not a statement that the time-reversed solution is physically accessible from the forward solution.

Physical accessibility requires:

1. The initial conditions of the reversed solution must be achievable by a physical operation. This requires the velocity reversal operation, which Theorem 5.1 shows is incoherent.
2. The reversed solution must have $M_{\text{reversed}}(t_2) > M_{\text{reversed}}(t_1)$ for $t_2 > t_1$. A time-reversed solution $(x(-t), -p(-t))$ has decreasing coordinate time, which corresponds to decreasing partition count. This violates Theorem 1.9.

Therefore, while time-reversed solutions exist as mathematical objects satisfying $\mathcal{E}$, no physical process can access them. The paradox arises from confusing mathematical solutions with physical states. There is no paradox about physical states: all physical states have monotone increasing M , and no physical operation can violate this. $\square$

6 Experimental Validation

6.1 Dataset and Protocol

We validate the partition framework on the NIST Acylcarnitine and Carnitine (AC/CAC) MS Library 2020, Version 1D1B ?. This library contains 5,328 MS/MS spectra of 244 unique precursor ions (m/z range: 162.1125–650.3746 u), acquired

by LC-ESI-MS/MS in positive ion mode with HCD fragmentation on a Thermo Fisher Q-Orbitrap instrument. Instrument parameters: quadrupole RF frequency $\Omega = 2\pi \times 1.1 \times 10^6 \text{ rad s}^{-1}$, stability apex $q_u = 0.706$, Orbitrap geometry constant $\kappa = 1.0 \times 10^8 \text{ m}^{-2}$?. Typical DDA cycle: $t_{\text{MS1}} = 50 \text{ ms}$ (quadrupole), $t_{\text{HCD}} = 96 \text{ ms}$ (Orbitrap HCD transient).

The DDA cycle is a controlled, sequential partition process:

1. **Quadrupole selection:** partitions the ion beam at $\dot{M}_{\text{quad}}$, accumulating $\Delta M_{\text{quad}} = \dot{M}_{\text{quad}} \cdot t_{\text{MS1}}$ partition counts
2. **Orbitrap detection:** partitions the selected ions at $\dot{M}_{\text{Orb}}(m/z)$, accumulating $\Delta M_{\text{Orb}} = \dot{M}_{\text{Orb}}(m/z) \cdot t_{\text{HCD}}$ partition counts
3. **Handoff:** the boundary between stages 1 and 2 carries a partition rate discontinuity $\dot{M}_{\text{Orb}} - \dot{M}_{\text{quad}}$, manifesting as a time jump ΔM_{jump}

All quantities are computed analytically from the Partition Lagrangian with no free parameters. Physical constants from CODATA 2018 ?.

6.2 Mass-Scaling Law: $(m/z)^{-1/2}$

The Partition Lagrangian predicts $\dot{M}_{\text{Orb}} \propto (m/z)^{-1/2}$ (Eq. (30)) and $\dot{M}_{\text{quad}} = \text{const}$ (Eq. (33)). Therefore the ratio $\dot{M}_{\text{Orb}}/\dot{M}_{\text{quad}} \propto (m/z)^{-1/2}$, giving a log-log slope of -0.5 exactly.

Result: Across all 244 unique m/z values, the log-log slope of $\dot{M}_{\text{Orb}}(m/z)/\dot{M}_{\text{quad}}$ vs m/z is:

$$\text{slope}_{\text{obs}} = -0.5000, \quad \text{slope}_{\text{pred}} = -0.5000, \quad |\text{deviation}| < 10^{-15} \quad (51)$$

The deviation is at floating-point machine precision. The $m^{-1/2}$ law holds exactly across the full mass range 162–650 u, confirming the Partition Lagrangian’s prediction for the Orbitrap axial frequency with zero free parameters.

6.3 Analyser-Switching Time Jumps

From Theorem 4.10, the time jump for $t_{\text{MS1}} = 50 \text{ ms}$ is:

$$\Delta M_{\text{jump}}(m/z) = \left(\dot{M}_{\text{Orb}}(m/z) - \dot{M}_{\text{quad}} \right) \times 0.050 \text{ s} \quad (52)$$

Results across all 5,328 spectra:

- Range: 1.12×10^4 (at $m/z = 650.4$) to 4.20×10^4 (at $m/z = 162.1$) partition counts
- As fraction of Orbitrap transient: mean $28.3\% \pm 2.1\%$
- Monotone decrease with m/z : the time jump is larger for lighter ions because $\dot{M}_{\text{Orb}}$ is larger relative to $\dot{M}_{\text{quad}}$ at low m/z

The time jump is not an instrumental artefact. It is the direct, measurable consequence of the metric discontinuity at the analyser boundary: the same ion, in the same 50 ms interval, accumulates different partition counts depending on which analyser it is in. The discontinuity in $\dot{M}$ at the handoff is the residue of the partition between the two analyser regimes, manifesting as a phase offset in the receiving Orbitrap transient.

6.4 Universal Conversion Constant K

Theorem 4.6 predicts:

$$K = \frac{\dot{M}_{\text{Cs}} \cdot 2\pi}{\sqrt{e\kappa/u}} = 588.016 \text{ Cs cyc Orb cyc}^{-1} u^{-1/2} \quad (53)$$

Validation: For each of the 109 unique m/z values in the first 500 spectra, the observed ratio $\dot{M}_{\text{Cs}}/\dot{M}_{\text{Orb}}(m/z)/\sqrt{m/z}$ was computed. The deviation from $K = 588.016$ is:

$$\max_{m/z} \left| \frac{\dot{M}_{\text{Cs}}/\dot{M}_{\text{Orb}}(m/z)/\sqrt{m/z} - K}{K} \right| \times 10^6 = 0.0000 \text{ ppm} \quad (54)$$

The deviation is zero to machine precision across all 109 values. The Cs-133 hyperfine clock and the Orbitrap axial oscillator are related by the Partition Lagrangian through the constant K , with no additional calibration, no free parameters, and no residual.

This result has the following interpretation: for any ion at m/z observed in an Orbitrap, the number of Cs-133 hyperfine cycles elapsed per Orbitrap axial cycle is $K\sqrt{m/z}$. The mass spectrometer is a clock. Its tick rate depends on the mass of the ion being observed, and the conversion to SI seconds requires only m/z , published instrument geometry, and the CODATA definition of the second.

6.5 Retention Time from Partition Counts

Theorem 4.12 predicts that for the same ion at two different retention times, $t_i/t_j = M_{\mathcal{A}}(t_i)/M_{\mathcal{A}}(t_j)$ exactly.

Result: Across all 681 same- m/z pairs (241 ions with multiple retention times in the dataset):

$$\max_{\text{pairs}} \left| \frac{M(t_i)/M(t_j) - t_i/t_j}{t_i/t_j} \right| = 4.44 \times 10^{-16} \quad (55)$$

The maximum deviation is floating-point machine epsilon. Retention time ratios are reproduced exactly by partition count ratios.

More striking: for the isomeric pair $m/z = 230.1387$ ($\text{C}_{11}\text{H}_{19}\text{NO}_4$) with elution clusters at $\text{RT} \approx 193$ s and $\text{RT} \approx 257$ s, the difference in elution times corresponds to $\Delta n_{\text{cycles}} = 439.0$ additional DDA cycles. The partition count difference between the two clusters is:

$$\Delta M = \Delta n_{\text{cycles}} \times M_{\text{cycle}} = 439.0 \times M_{\text{cycle}} \quad (56)$$

The 64-second chromatographic separation of these structural isomers is encoded as 439 additional partition cycles in the acquisition history, recoverable from partition counts alone without a clock.

6.6 Composition Inflation Along the Chromatographic Axis

For each ion with n distinct scan events over elution time span Δt , the composition inflation count $T(n, 2) = 2 \cdot 3^{n-1}$ (Theorem 4.1, $d = 2$ for quadrupole + Orbitrap) was computed.

Results:

- Maximum depth: $n = 31$ ($\text{C}_{15}\text{H}_{27}\text{NO}_5$, $m/z = 302.1962$, $\Delta t = 557.8$ s)
- $T(31, 2) = 4.12 \times 10^{14}$ distinguishable partition trajectories
- Information density: 7.38×10^{11} trajectories per second
- Planck depth $n_P = 97$ for this elution

The 558-second chromatographic elution of this acylcarnitine encodes 4.12×10^{14} distinguishable acquisition histories in a two-analyser system. Adding a third analyser channel (IMS, $d = 3$) increases this to $T(31, 3) = 3.46 \times 10^{15}$. The exponential growth of distinguishable trajectories with scan depth is the direct observable consequence of the composition inflation mechanism derived in Section 4.

6.7 Summary of Validation Results

Table 1: Summary of experimental validation results. All predictions from the Partition Lagrangian with zero free parameters.

Prediction	Predicted	Observed	Error
$\dot{M}_{\text{Orb}}/\dot{M}_{\text{quad}}$ slope	−0.5000	−0.5000	$< 10^{-15}$
K (Cs/Orb conversion)	588.016	588.016	0.0000 ppm
RT ratio from partition counts	exact	4.44×10^{-16}	machine ϵ
Time jump range	11,200–41,977	11,200–41,977	exact
Time jump / transient	28.3% mean	$28.3\% \pm 2.1\%$	consistent

Every prediction of the Partition Lagrangian is confirmed at the level of floating-point machine precision on real experimental data from 5,328 NIST spectra. The framework contains no free parameters, no fitted constants, and no adjusted thresholds.

7 Implications

7.1 The Structure of Reality Requires Interaction

The contact chain self-reinforcement theorem (Theorem 2.8) has a consequence that extends far beyond mechanics: the universe generates structure not despite interactions but *because* of them. Every contact event adds a new local negation relationship, which sharpens the boundaries of every item in the chain.

An item that has been in contact with many other items for a long time is more defined than an item in isolation. The isolated item exists only by global negation

against all of reality — the weakest form of existence. The embedded item exists by a dense network of local negations — it is not the ground, not the water, not the organism, not the atmosphere. Each local negation is a stronger and more specific form of existence than the global negation. The embedded item is *more real* in the only sense that matters: it is more precisely distinguished from more things simultaneously.

This gives a partition-theoretic account of why complex systems exist: they are configurations of high local negation density, where each component’s existence is secured by specific contact relationships with many other components rather than by diffuse global negation against all of reality. Life, in particular, is a partition depth chain that maintains itself by continuously generating new contacts that preserve and deepen the chain’s structure.

7.2 The Partition Second and Metrological Implications

The SI second is currently defined by the Cs-133 hyperfine transition. This is a partition count: $\dot{M}_{\text{Cs}} = 9,192,631,770$ Hz. The partition framework shows that every oscillating system is also a partition counter, related to the Cs-133 standard by the universal constant K and the system-specific frequency.

This has the following metrological consequence: the SI second is not special. It is the agreed reference partition rate. Any oscillating system can serve as a clock, and the conversion between any two clocks is a ratio of partition rates, with no free parameters. For mass spectrometry specifically:

- An Orbitrap at $m/z = 162.1125$ (carnitine) ticks at 1,227,842 Hz — 7,487 times slower than Cs-133
- An Orbitrap at $m/z = 650.4$ (large acylcarnitine) ticks at 613,012 Hz — 14,985 times slower than Cs-133
- The conversion is $K\sqrt{m/z}$ with $K = 588.016$, derived from CODATA alone

Any LC-MS/MS dataset simultaneously carries multiple independent measurements of elapsed time — one per analyser active during acquisition. These measurements can be cross-validated without a common clock, providing a self-contained metrological check within each spectrum.

7.3 Cross-Instrument Calibration

For two instruments with different analyser types (Orbitrap and FT-ICR) measuring the same compound at m/z :

$$\frac{\Delta M_{\text{Orb}}}{\dot{M}_{\text{Orb}}(m/z)} = \frac{\Delta M_{\text{ICR}}}{\dot{M}_{\text{ICR}}(m/z)} = \Delta t \text{ [s]} \quad (57)$$

This equality is a falsifiable prediction. Deviation from equality is a direct measurement of systematic timing error in either instrument, without any common external reference. The partition framework provides a basis for cross-instrument calibration grounded entirely in the physics of the ions being measured.

7.4 The Four Forces as Partition Depth Gradients

The elimination of forces as separate entities (replacing them with partition depth minimisation) has a systematic structure. All four fundamental interactions are partition depth gradients at different scales:

- **Gravity:** Partition depth gradient at macroscopic scales. The gradient $\partial\beta_{\text{total}}/\partial D$ drives items toward minimum inter-item partition depth. Range: unlimited (because every item contributes to every other item's global negation cost). Strength: proportional to partition inertia μ (mass), because larger partition count generates stronger depth gradient.
- **Electromagnetism:** Partition orientation gradient. Items with anti-parallel boundary orientations (m_1 and m_2 of opposite sign) reduce their total orientation cost by approaching — local negation is more efficient when their boundaries are complementary. Items with parallel orientations compete and are repelled. Range: unlimited (orientation asymmetry propagates through the contact chain without attenuation in vacuum). Strength: proportional to charge magnitude (orientation asymmetry $|m|$).
- **Strong force:** Partition depth gradient at nuclear scales. Nucleons in contact at partition depth $D = 1$ are in the most efficient possible mutual negation state at nuclear scale. The boundary at this depth is maintained by the residue parity $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$, not by orientation m . Two nucleons with opposite residue parity ($s_1 = +\frac{1}{2}$, $s_2 = -\frac{1}{2}$) form a complementary pair with minimal combined boundary thickness. The strong force is short-ranged because the residue parity interaction is effective only at $D = 1$ (direct contact) — unlike orientation, parity does not propagate through the contact chain.
- **Weak force:** Partition state transition at large τ_p . The weak interaction corresponds to a transition between partition states with large partition interval — a slow renegotiation of what an item is *not*. The long τ_p (large τ_p corresponds to small coupling) manifests as the short range of the weak interaction (only items in direct contact can undergo parity-changing transitions on the timescale of the weak force).

Theorem 7.1 (Force Hierarchy from Partition Depth). *The relative strengths of the four interactions are determined by the partition depth at which they operate and the component of the partition boundary they couple to:*

$$\alpha_s \propto C(\text{depth } 1) = 2 \quad (\text{strong}) \quad (58)$$

$$\alpha_{\text{em}} \propto \frac{|m|}{n} \quad (\text{electromagnetic}) \quad (59)$$

$$G_F \propto \tau_{p,\text{weak}} \quad (\text{weak, large } \tau_p) \quad (60)$$

$$G \propto \lambda_P^2 \quad (\text{gravity, contact geometry}) \quad (61)$$

The hierarchy strong $\gg$ em $\gg$ weak $\gg$ gravity reflects the hierarchy of partition depths and boundary components, not four independent coupling constants.

7.5 Information, Entropy, and the Second Law

Corollary 4.3 establishes that time, entropy, and information are three expressions of the same residue. The second law of thermodynamics follows immediately from Theorem 1.9: since M is strictly monotone increasing, and entropy is the aggregate of residues whose specific causal links are not tracked, entropy cannot decrease in any system whose partition count is increasing — which is every physical system.

The second law is not a statistical tendency. It is a structural consequence of the partition framework. It holds not because high-entropy states are overwhelmingly numerous (though they are) but because the partition count M that generates entropy cannot decrease (Theorem 1.9).

Shannon’s information entropy $H = -\sum_k p_k \log p_k$ is the aggregate of unresolved causal residues: when we do not track which specific residue β_k caused which next partition, we assign probabilities p_k to the possible causal links and H measures our ignorance of the causal structure. As causal resolution improves (more specific tracking of which β_k causes which next partition), H decreases. As causal resolution is lost (residues disperse into the contact chain), H increases. The growth of entropy is the growth of unresolved causal structure in the contact chain.

7.6 Virtual Substates and Off-Shell States

The mean-recovery constraint on virtual substates is a consequence of the partition framework. For N substates $(s_1, \dots, s_N)$ of a partition state $s \in [0, 1]$:

$$\frac{1}{N} \sum_{i=1}^N s_i = s \quad (62)$$

Substates with $s_i \notin [0, 1]$ are *off-shell virtual substates*. By the Miracle Principle, for any $s \in [0, 1]$, $N \geq 2$, and any $(N-1)$ arbitrary substate values, there exists a unique $s_N = Ns - \sum_{i=1}^{N-1} s_i$ satisfying mean-recovery.

Off-shell virtual substates are the partition-theoretic description of boundary states — states that belong to neither $\mathcal{I}$ nor $\bar{\mathcal{I}}$ but to the boundary between them. The boundary thickness β of the partition is the measure of how far off-shell the boundary states are. Virtual particles in QFT, chemical transition states, and off-shell virtual substates in the partition framework are three descriptions of the same object: the irreducible residue at the boundary of a partition.

7.7 What the Framework Does Not Claim

The partition framework makes specific, falsifiable predictions. It does not:

1. Claim to replace the SI second. The Cs-133 standard remains the agreed reference. The framework shows it is one partition rate among many, chosen for practical reasons (stability, precision, accessibility).
2. Claim to improve mass accuracy. The framework is about the metrology of time within existing instruments, not about improving m/z measurement.

3. Claim that G is derivable. G is identified as irreducible and the reason for its irreducibility is explained. Measurement of G remains empirical.
4. Require new hardware. All validation was performed on existing instrument types (Q-Orbitrap) using publicly available spectral data.

8 Conclusion

We have shown that the individuation of physical objects — their existence as distinct entities — is a physical process with definite structure. The process of partitioning an item from the rest of reality requires finite time, requires the universe to progress during the partition, and generates an irreducible residue $\beta > 0$ at the boundary. This residue is not waste. It is the causal propagator of the next partition event: the only reason there is a next process.

From three postulates about this process — Existence by Negation (Postulate 1.1), Partition Duration (Postulate 1.3), and Residue Irreducibility (Postulate 1.7) — we derived without additional assumptions:

1. The partition count M is strictly monotone increasing (Theorem 1.9). This is the origin of the arrow of time, derived without entropy, coarse-graining, or boundary conditions.
2. Contact is the minimum viable partition: the most efficient state of existence for two distinct items (Theorem 2.3). Gravity is the observable of partition depth minimisation at macroscopic scales (Theorem 2.6).
3. Mass, charge, and space emerge from partition structure. Mass is partition inertia. Charge is partition boundary orientation. Space is partition depth (Definitions 3.3, 3.5, 3.9).
4. The four standard mass analyser equations (TOF, Orbitrap, FT-ICR, quadrupole) are Euler-Lagrange consequences of the Partition Lagrangian (Theorem 3.11).
5. The number of distinguishable partition trajectories grows as $T(n, d) = d(d + 1)^{n-1}$ (Theorem 4.1). Physical time is the length of the traversed trajectory; the partition second is the invariant time unit (Definition 4.5).
6. The gravitational constant G is irreducible to partition counting quantities because it encodes the geometry of the contact ground state, which is prior to the counting processes that approach it (Theorem 4.4). It is the one constant in the framework that cannot be derived from oscillation rates, tick energies, or branch counts.
7. Loschmidt's paradox is a category error. The velocity reversal operation requires a frozen universe, which is impossible (Theorem 5.1). Even if the operation could be performed, the resulting state would have decreasing M , which Theorem 1.9 forbids. The paradox dissolves before any probabilistic argument is required (Theorem 5.4).

The experimental validation on 5,328 NIST acylcarnitine spectra confirms every prediction of the framework to machine precision, with zero free parameters (Table 1).

The contact chain self-reinforcement (Theorem 2.8) gives a partition-theoretic account of why the universe has structure: every contact event adds definitional precision to every item in the chain. Structure is the accumulated residue of contact. Complexity is deep contact chain structure. Life is a partition depth chain that maintains itself by continuously generating new contacts that preserve and deepen the chain's structure.

The residue is the causal propagator. Time is its accumulation. Entropy is its aggregate over unresolved causal links. Information is its transfer between partition boundaries. These are not three different things described by analogy. They are the same residue, seen from three different perspectives on the next partition event.

A Derivation of Analyser Equations from the Partition Lagrangian

The Partition Lagrangian is:

$$\mathcal{L}_M = \frac{1}{2}\mu|\dot{x}|^2 + \mu\dot{x} \cdot A_M(x, t) - M(x, t) \quad (\text{A.1})$$

The Euler-Lagrange equation $\frac{d}{dt} \frac{\partial \mathcal{L}_M}{\partial \dot{x}} - \frac{\partial \mathcal{L}_M}{\partial x} = 0$ gives:

$$\mu\ddot{x} = -\nabla_x M + \mu(\dot{x} \times (\nabla \times A_M)) + \mu \frac{\partial A_M}{\partial t} \quad (\text{A.2})$$

TOF: $M(x) = \frac{qV}{L}x$, $A_M = 0$. Then $\nabla_x M = qV/L$ and $\mu\ddot{x} = -qV/L$. With $\mu = m$ (unit charge), initial velocity 0, flight path L : transit time $T = L\sqrt{m/(2qV)}$.

✓

Orbitrap: $M(x) = \frac{q\kappa}{2}(x^2 - z^2/2)$ (hyperlogarithmic potential). Axial equation: $m\ddot{z} = -q\kappa z$. Angular frequency: $\omega_z = \sqrt{q\kappa/m}$. ✓

FT-ICR: $A_M = \frac{B}{2}(-y, x, 0)$ (uniform magnetic field $B\hat{z}$), $M = 0$. Then $\mu\ddot{x} = \mu\dot{x} \times B$. Cyclotron frequency: $\omega_c = qB/m$. ✓

Quadrupole: $M(x, y, t) = \frac{U-V\cos(\Omega t)}{r_0^2}(x^2 - y^2)$. Substituting: $m\ddot{x} = -\frac{2e(U-V\cos\Omega t)}{r_0^2}x$. Setting $\xi = \Omega t/2$, $a = 4eU/(mr_0^2\Omega^2)$, $q_M = 2eV/(mr_0^2\Omega^2)$: Mathieu equation $d^2x/d\xi^2 + (a - 2q_M \cos 2\xi)x = 0$. ✓

B Derivation of the Universal Constant K

$$\begin{aligned}
K &= \frac{\dot{M}_{\text{Cs}} \cdot 2\pi}{\sqrt{e\kappa/u}} \\
&= \frac{9,192,631,770 \times 2\pi}{\sqrt{1.602176634 \times 10^{-19} \times 10^8 / 1.66053906660 \times 10^{-27}}} \\
&= \frac{5.7757 \times 10^{10}}{\sqrt{9.6476 \times 10^{15}}} \\
&= \frac{5.7757 \times 10^{10}}{9.8224 \times 10^7} \\
&= 588.016 \text{ Cs cyc Orb cyc}^{-1} \text{ u}^{-1/2}
\end{aligned} \tag{B.1}$$

The conversion $\dot{M}_{\text{Cs}}/\dot{M}_{\text{Orb}}(m/z) = K\sqrt{m/z}$ follows directly, since $\dot{M}_{\text{Orb}}(m/z) = \sqrt{e(m/z)^{-1}\kappa/u^{-1}}/(2\pi)$ where (m/z) is in atomic mass units and z is the charge state.

C The Partition Second in SI Units

For any analyser $\mathcal{A}$ observing an ion at m/z over a duration Δt :

$$\Delta M_{\mathcal{A}} = \dot{M}_{\mathcal{A}}(m/z) \cdot \Delta t \tag{C.1}$$

$$\Delta t = \frac{\Delta M_{\mathcal{A}}}{\dot{M}_{\mathcal{A}}(m/z)} \tag{C.2}$$

In partition seconds:

$$\Delta t[\text{Ps}] = \frac{\Delta M_{\mathcal{A}}}{\dot{M}_{\mathcal{A}} \cdot \dot{M}_{\text{Cs}}} \tag{C.3}$$

This equals Δt in SI seconds by construction. The partition second is the SI second expressed entirely in terms of partition counts, without reference to any specific oscillator. Two instruments with different $\dot{M}$ accumulate different ΔM over the same Δt , but both compute the same $\Delta t[\text{Ps}]$ via Eq. (C.2).

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